

SHIV NADAR

— UNIVERSITY —

CHENNAI

TOOLS FOR DATA VISUALIZATION

ASSIGNMENT 12



EXPLORE



ANALYZE



VISUALIZE



INSIGHT



SHIV NADAR UNIVERSITY
CHENNAI



25011101003



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Experiment No. 12

Dashboard with KPI Indicators and Cards Using Power BI

“Without data, you’re just another person with an opinion.”

— W. Edwards Deming

Introduction

Dashboards are powerful tools that provide a snapshot of business performance by visually representing key metrics. Power BI enables creating interactive dashboards with KPI (Key Performance Indicator) indicators and cards that highlight important data points such as sales performance, customer growth, or inventory status. These visual elements help stakeholders make informed decisions quickly by focusing on the most critical numbers.

This experiment introduces the creation of KPI indicators and card visuals in Power BI to summarize and track essential business metrics dynamically.

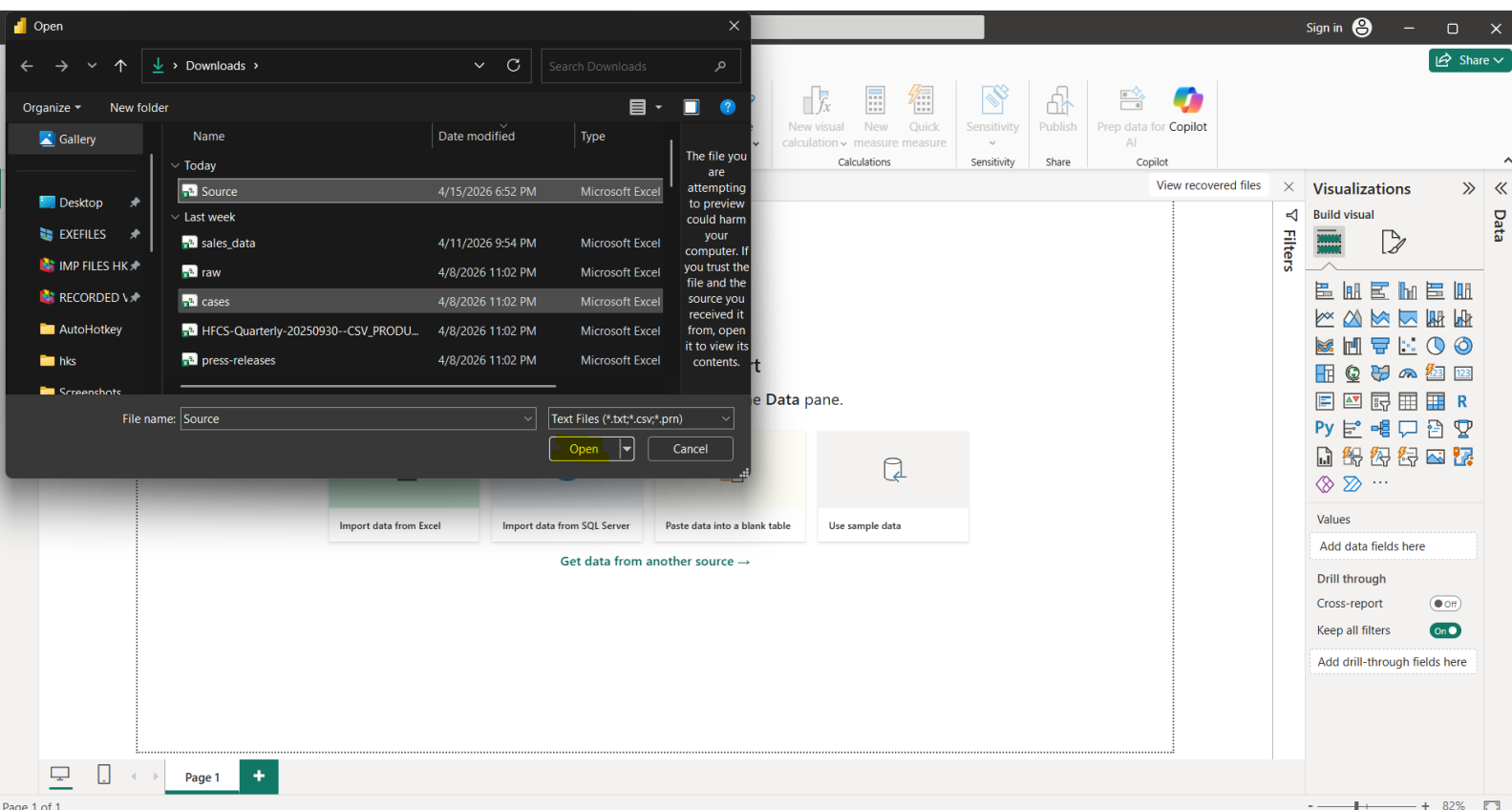
Aim:

To design an interactive dashboard in Power BI using KPI indicators and card visuals to effectively display and monitor key business performance metrics.

Objective

To create a dashboard in Power BI featuring KPI indicators and cards that display key business metrics clearly and effectively.

STEPS TO MAKE DASHBOARD



https://app.powerbi.com/links/axEGJIm8ma?ctid=5beb351c-3fb8-418f-b612-fe36ace96ef3&pbi_source=linkShare

FileHomeTransformAdd ColumnViewToolsHelp

Close & ApplyClose

New SourceNew Query

Recent Sources

Enter Data

Data source settingsData Sources

Manage ParametersParameters

Export query resultsOutput Data

Refresh Preview

PropertiesAdvanced EditorManageQuery

Choose ColumnsRemove ColumnsManage Columns

Keep RowsRemove RowsReduce Rows

Sort

Split ColumnGroup ByTransform

Use First Row as HeadersReplace Values

Merge QueriesAppend QueriesCombine FilesCombine

Queryes [1]

Copy

Paste

Delete

Rename

Enable load

Include in report refresh

Duplicate

Reference

Move To Group

Move Up

Move Down

Create Function...

Convert To Parameter

Advanced Editor

Properties...

Index

Runs

Opponent

Ground

Date

Match

Match_No

0	12	SriLanka	Dambulla	8/18/2008	ODI	
1	37	SriLanka	Dambulla	8/20/2008	ODI	
2	25	SriLanka	Colombo(RPS)	8/24/2008	ODI	
3	54	SriLanka	Colombo(RPS)	8/27/2008	ODI	
4	31	SriLanka	Colombo(RPS)	8/29/2008	ODI	
5	2	SriLanka	Colombo(RPS)	9/14/2009	ODI	
6	16	Pakistan	Centurion	9/26/2009	ODI	
8	79	Westindies	Johannesburg	9/30/2009	ODI	
9	30	Australia	Vadodara	10/25/2009	ODI	
10	10	Australia	Mohali	11/2/2009	ODI	
11	27	SriLanka	Rajkot	12/15/2009	ODI	
12	54	SriLanka	Nagpur	12/18/2009	ODI	
13	107	SriLanka	Kolkata	12/24/2009	ODI	
15	9	SriLanka	Mirpur	1/5/2010	ODI	
16	91	Bangladesh	Mirpur	1/7/2010	ODI	
17	71	SriLanka	Mirpur	1/10/2010	ODI	
17	102	Bangladesh	Mirpur	1/11/2010	ODI	
18	19	2	Mirpur	1/13/2010	ODI	
19	20	31	Jaipur	2/21/2010	ODI	
20	57	SouthAfrica	Ahmedabad	2/27/2010	ODI	
21	23	0	Bulawayo	5/28/2010	ODI	
22	24	82	Bulawayo	5/30/2010	ODI	
23	25	18	Harare	6/3/2010	ODI	
24	26	68	Harare	6/5/2010	ODI	
25	27	26	Harare	6/12/2010	T20	
26	29	11	Dambulla	6/16/2010	ODI	
27	30	18	Dambulla	6/19/2010	ODI	
28						

Query Settings

PROPERTIES

Name

VIRAT KHOLI

All Properties

APPLIED STEPS

Source

Promoted Headers

Changed Type

8 COLUMNS, 516 ROWSColumn profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 6:55 PM

FileHomeTransformAdd ColumnViewToolsHelp

Close & ApplyClose

New SourceNew Query

Recent SourcesEnter Data

Data source settingsData Sources

Manage ParametersParameters

Export query resultsOutput Data

Refresh PreviewQuery

PropertiesAdvanced EditorManage

Choose ColumnsRemove ColumnsManage Columns

Keep RowsRemove RowsReduce Rows

Sort

Split ColumnGroup ByTransform

12Replace Values

Merge QueriesAppend QueriesCombine Files

Combine

Queries [1]

VIRAT KHOLI

fx

= Table.TransformColumnTypes(#"Promoted Headers",{{"index", Int64.Type}, {"runs", Int64.Type}, {"opponent", type text}, {"ground", type

index	runs	opponent	ground	date	match	Match_No
1	0	12 SriLanka	Dambulla	8/18/2008	ODI	
2	1	37 SriLanka	Dambulla	8/20/2008	ODI	
3	2	25 SriLanka	Colombo(RPS)	8/24/2008	ODI	
4	3	54 SriLanka	Colombo(RPS)	8/27/2008	ODI	
5	4	31 SriLanka	Colombo(RPS)	8/29/2008	ODI	
6	5	2 SriLanka	Colombo(RPS)	9/14/2009	ODI	
7	6	16 Pakistan	Centurion	9/26/2009	ODI	
8	8	79 Westindies	Johannesburg	9/30/2009	ODI	
9	9	30 Australia	Vadodara	10/25/2009	ODI	
10	10	10 Australia	Mohali	11/2/2009	ODI	
11	11	27 SriLanka	Rajkot	12/15/2009	ODI	
12	12	54 SriLanka	Nagpur	12/18/2009	ODI	
13	13	107 SriLanka	Kolkata	12/24/2009	ODI	
14	15	9 SriLanka	Mirpur	1/5/2010	ODI	
15	16	91 Bangladesh	Mirpur	1/7/2010	ODI	
16	17	71 SriLanka	Mirpur	1/10/2010	ODI	
17	18	102 Bangladesh	Mirpur	1/11/2010	ODI	
18	19	2 SriLanka	Mirpur	1/13/2010	ODI	
19	20	31 SouthAfrica	Jaipur	2/21/2010	ODI	
20	22	57 SouthAfrica	Ahmedabad	2/27/2010	ODI	
21	23	0 Zimbabwe	Bulawayo	5/28/2010	ODI	
22	24	82 SriLanka	Bulawayo	5/30/2010	ODI	
23	25	18 Zimbabwe	Harare	6/3/2010	ODI	
24	26	68 SriLanka	Harare	6/5/2010	ODI	
25	27	26 Zimbabwe	Harare	6/12/2010	T20	
26	29	11 Bangladesh	Dambulla	6/16/2010	ODI	
27	30	18 Pakistan	Dambulla	6/19/2010	ODI	
28						

Query Settings

PROPERTIES

Name

VIRAT KHOLI

All Properties

APPLIED STEPS

Source

Promoted Headers

Changed Type

8 COLUMNS, 516 ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 6:55 PM

FileHomeHelpTable toolsColumn tools

NameColumn

Data typeWhole number

Format

SummarizationSum

Data categoryUncategorized

Sort by column

Data groups

Manage relationships

New column

Auto recovery contains some recovered files that haven't been opened.

1 100s =
2 VAR res = SWITCH(TRUE(),
3 'Virat_Kohli'[runs]>=100 && 'Virat_Kohli'[runs]<200, 1,
4 'Virat_Kohli'[runs]>=200 && 'Virat_Kohli'[runs]<300, 2,
5 'Virat_Kohli'[runs]>=300 && 'Virat_Kohli'[runs]<400, 3,
6 0)
7 RETURN res
8

index	runs	opponent	ground	date	match	Match_No	total	Column
0	12	SriLanka	Dambulla	Monday, August 18, 2008	ODI	1	12	
1	37	SriLanka	Dambulla	Wednesday, August 20, 2008	ODI	2	49	
2	25	SriLanka	Colombo(RPS)	Sunday, August 24, 2008	ODI	3	74	
3	54	SriLanka	Colombo(RPS)	Wednesday, August 27, 2008	ODI	4	128	
4	31	SriLanka	Colombo(RPS)	Friday, August 29, 2008	ODI	5	159	
5	2	SriLanka	Colombo(RPS)	Monday, September 14, 2009	ODI	6	161	
6	16	Pakistan	Centurion	Saturday, September 26, 2009	ODI	7	177	
8	79	WestIndies	Johannesburg	Wednesday, September 30, 2009	ODI	8	256	
9	30	Australia	Vadodara	Sunday, October 25, 2009	ODI	9	286	
10	10	Australia	Mohali	Monday, November 2, 2009	ODI	10	296	
11	27	SriLanka	Rajkot	Tuesday, December 15, 2009	ODI	11	323	
12	54	SriLanka	Nagpur	Friday, December 18, 2009	ODI	12	377	
13	107	SriLanka	Kolkata	Thursday, December 24, 2009	ODI	13	484	
15	9	SriLanka	Mirpur	Tuesday, January 5, 2010	ODI	14	493	
16	91	Bangladesh	Mirpur	Thursday, January 7, 2010	ODI	15	584	
17	71	SriLanka	Mirpur	Sunday, January 10, 2010	ODI	16	655	
18	102	Bangladesh	Mirpur	Monday, January 11, 2010	ODI	17	757	
19	2	SriLanka	Mirpur	Wednesday, January 13, 2010	ODI	18	759	
20	31	SouthAfrica	Jaipur	Sunday, February 21, 2010	ODI	19	790	
22	57	SouthAfrica	Ahmedabad	Saturday, February 27, 2010	ODI	20	847	

Virat_Kohli
Column
date
ground
index
match
Match_No
opponent
runs
total

Table: Virat_Kohli (516 rows) Column: Column (1 distinct values)

FileHomeHelpTable toolsColumn tools

NameCalendar

Manage relationships

New measureQuick measure

New columnNew table

Mark as date table

StructureRelationshipsCalculationsCalendars

Auto recovery contains some recovered files that haven't been opened.

1 year = YEAR(Calendar[Date])

Date	day	day_no	month	month_no	year
1/1/2008 12:00:00 AM	Tue	1	Jan	1	2008
1/2/2008 12:00:00 AM	Wed	2	Jan	1	2008
1/3/2008 12:00:00 AM	Thu	3	Jan	1	2008
1/4/2008 12:00:00 AM	Fri	4	Jan	1	2008
1/5/2008 12:00:00 AM	Sat	5	Jan	1	2008
1/6/2008 12:00:00 AM	Sun	6	Jan	1	2008
1/7/2008 12:00:00 AM	Mon	7	Jan	1	2008
1/8/2008 12:00:00 AM	Tue	8	Jan	1	2008
1/9/2008 12:00:00 AM	Wed	9	Jan	1	2008
1/10/2008 12:00:00 AM	Thu	10	Jan	1	2008
1/11/2008 12:00:00 AM	Fri	11	Jan	1	2008
1/12/2008 12:00:00 AM	Sat	12	Jan	1	2008
1/13/2008 12:00:00 AM	Sun	13	Jan	1	2008
1/14/2008 12:00:00 AM	Mon	14	Jan	1	2008
1/15/2008 12:00:00 AM	Tue	15	Jan	1	2008
1/16/2008 12:00:00 AM	Wed	16	Jan	1	2008
1/17/2008 12:00:00 AM	Thu	17	Jan	1	2008
1/18/2008 12:00:00 AM	Fri	18	Jan	1	2008
1/19/2008 12:00:00 AM	Sat	19	Jan	1	2008
1/20/2008 12:00:00 AM	Sun	20	Jan	1	2008
1/21/2008 12:00:00 AM	Mon	21	Jan	1	2008
1/22/2008 12:00:00 AM	Tue	22	Jan	1	2008
1/23/2008 12:00:00 AM	Wed	23	Jan	1	2008
1/24/2008 12:00:00 AM	Thu	24	Jan	1	2008
1/25/2008 12:00:00 AM	Fri	25	Jan	1	2008
1/26/2008 12:00:00 AM	Sat	26	Jan	1	2008

View recovered files

Data

Search

Calendar

Date

day

day_no

month

month_no

year

Virat_Kohli

100s

30+

50s

date

ground

index

match

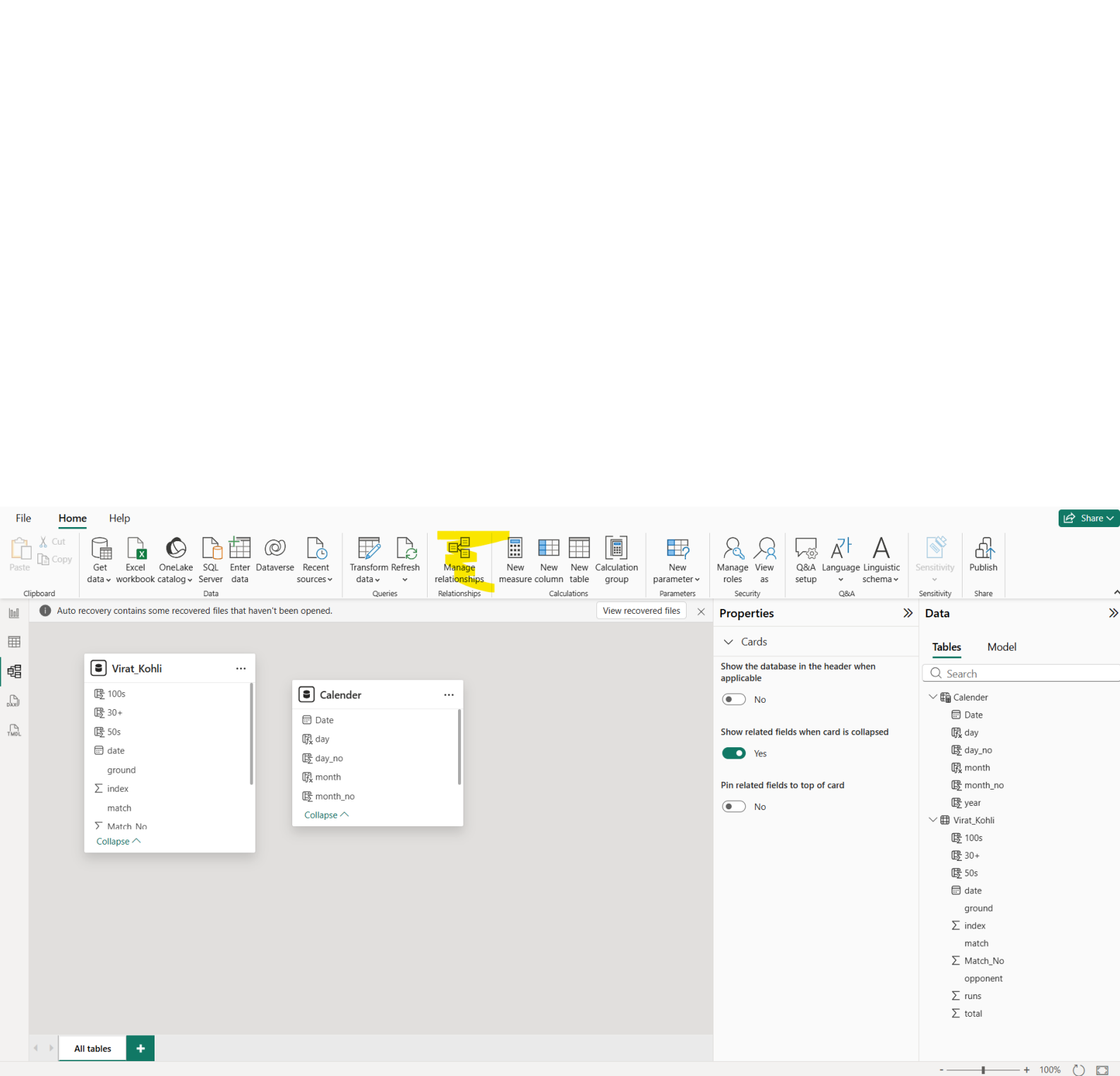
Match_No

opponent

runs

total

Table: Calendar (5,479 rows) Column: year (15 distinct values)



Manage relationships



+ New relationship

⚡ Autodetect

✎ Edit

🗑 Delete

≡ Filter ▾

☐ From: table (column) ↑

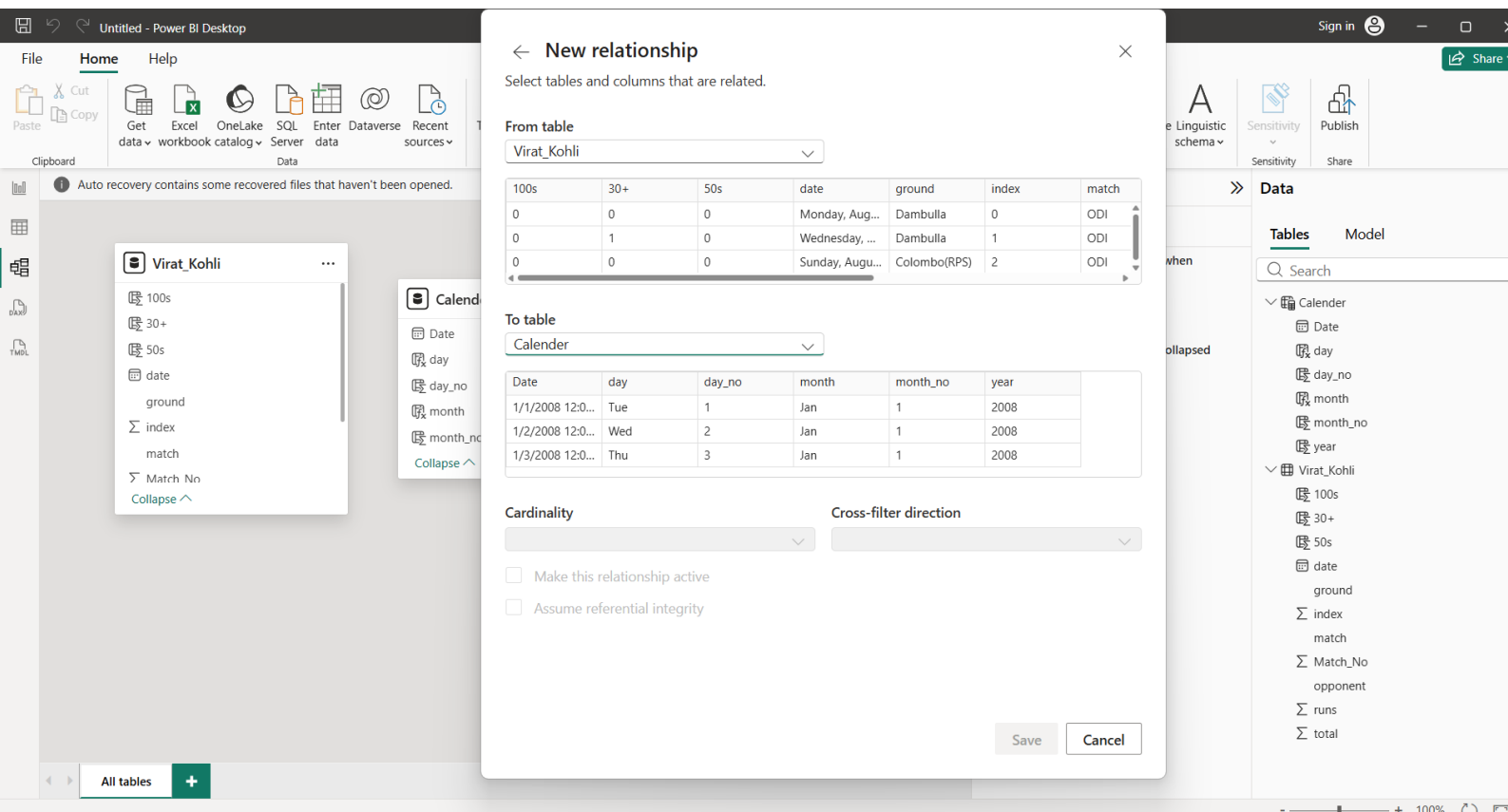
Relationship

To: table (column)

Status

There are no relationships defined yet.

Close



Untitled - Power BI Desktop

FileHomeHelp

PasteCutCopy

Get dataExcelOneLakeSQL ServerEnter dataDatawarehouseRecent sources

Auto recovery contains some recovered files that haven't been opened.

Virat_Kohli

100s30+50sdategroundindexmatch

Calendar

Date

day

day_no

month

month_no

year

100s30+50sdategroundindexmatch

000Monday, Aug...Dambulla0ODI

010Wednesday, A...Dambulla1ODI

000Sunday, Augu...Colombo(RPS)2ODI

Virat_Kohli

100s30+50sdategroundindexmatch

Calendar

Date

day

day_no

month

month_no

year

100s30+50sdategroundindexmatch

000Monday, Aug...Dambulla0ODI

010Wednesday, A...Dambulla1ODI

000Sunday, Augu...Colombo(RPS)2ODI

Cardinality

Many to one (*:1)

Cross-filter direction

Single

☒ Make this relationship active

☐ Assume referential integrity

☐ Apply security filter in both directions

SaveCancel

TablesModel

Search

Calendar

Date

day

day_no

month

month_no

year

Virat_Kohli

100s

30+

50s

date

ground

index

match

Match_No

opponent

runs

total



FileHomeHelp

PasteCutCopy

Get dataExcelOneLakeSQL ServerEnter dataDataaverseRecent sourcesTransform dataRefresh dataQueriesManage relationshipsNew measureNew columnNew tableCalculation groupNew parameterParametersManage rolesView asSecurityQ&A setupLanguage Linguistic schemaQ&ASensitivityPublish

Auto recovery contains some recovered files that haven't been opened.View recovered files

Virat_Kohli

100s30+50sdategroundindexmatchMatch NoCollapse

Calender

Datedayday_nomonthmonth_noCollapse

Relationship

TableColumn

Virat_Kohli

date

Cardinality

Many to one (*:1)

TableColumn

Calender

Date

Make this relationship active

Yes

Cross-filter direction

Single

Apply security filter in both directions

No

Apply changes

Open relationship editor

TablesModel

Search

Calender

Datedayday_nomonthmonth_noyear

Virat_Kohli

100s30+50sdategroundindexmatchMatch_Noopponentruns

total

All tables

+

100%100%

FileHomeHelpTable tools

NameTable

Manage relationships

New measure

Quick measure

New column

New table

Mark as date table

StructureRelationshipsCalculationsCalendars

Auto recovery contains some recovered files that haven't been opened.

View recovered files

1 Measures_Table=

Column

Search

Calendar

Date

day

day_no

month

month_no

year

Table

Virat_Kohli

100s

30+

50s

date

ground

index

match

Match_No

opponent

runs

total

Table: Table (1 rows)



File

Home

Help

Table tools

Measure tools

Name

100s

Format

Whole number

Data category

Uncategorized

New measure

Quick measure

Home table

Measures_Table

\$ %

0

Structure

Formatting

Properties

Calculations

View recovered files

Share

Auto recovery contains some recovered files that haven't been opened.

1 100s = SUM(Virat_Kohli[100s])

Column

Search

Calendar

Date

day

day_no

month

month_no

year

Measures_Table

runs

total

New measure

New column

New quick measure

Refresh

Manage relationships

Copy table

Rename

Delete from model

Hide in report view

Unhide all

Collapse all

Expand all

Table: Measures_Table (1 rows) Column: 100s (0 distinct values)

Table: Measures_Table (1 rows) Column: 100s (0 distinct values)

Untitled - Power BI Desktop

Search

Sign in

Share

FileHomeHelpTable toolsMeasure tools

NameTotal_Matches

FormatWhole number

Data categoryUncategorized

New Quick measure measure

Home tableMeasures_Table

\$ % 0

0

StructureFormattingPropertiesCalculations

Auto recovery contains some recovered files that haven't been opened.

View recovered files

1 Total_Matches = COUNT(Virat_Kohli[opponent])

Column

Search

Calender

Date

day

day_no

month

month_no

year

Measures_Table

100s

30+

50s

Column

HighestScore

Runs

Total_Matches

Virat_Kohli

100s

30+

50s

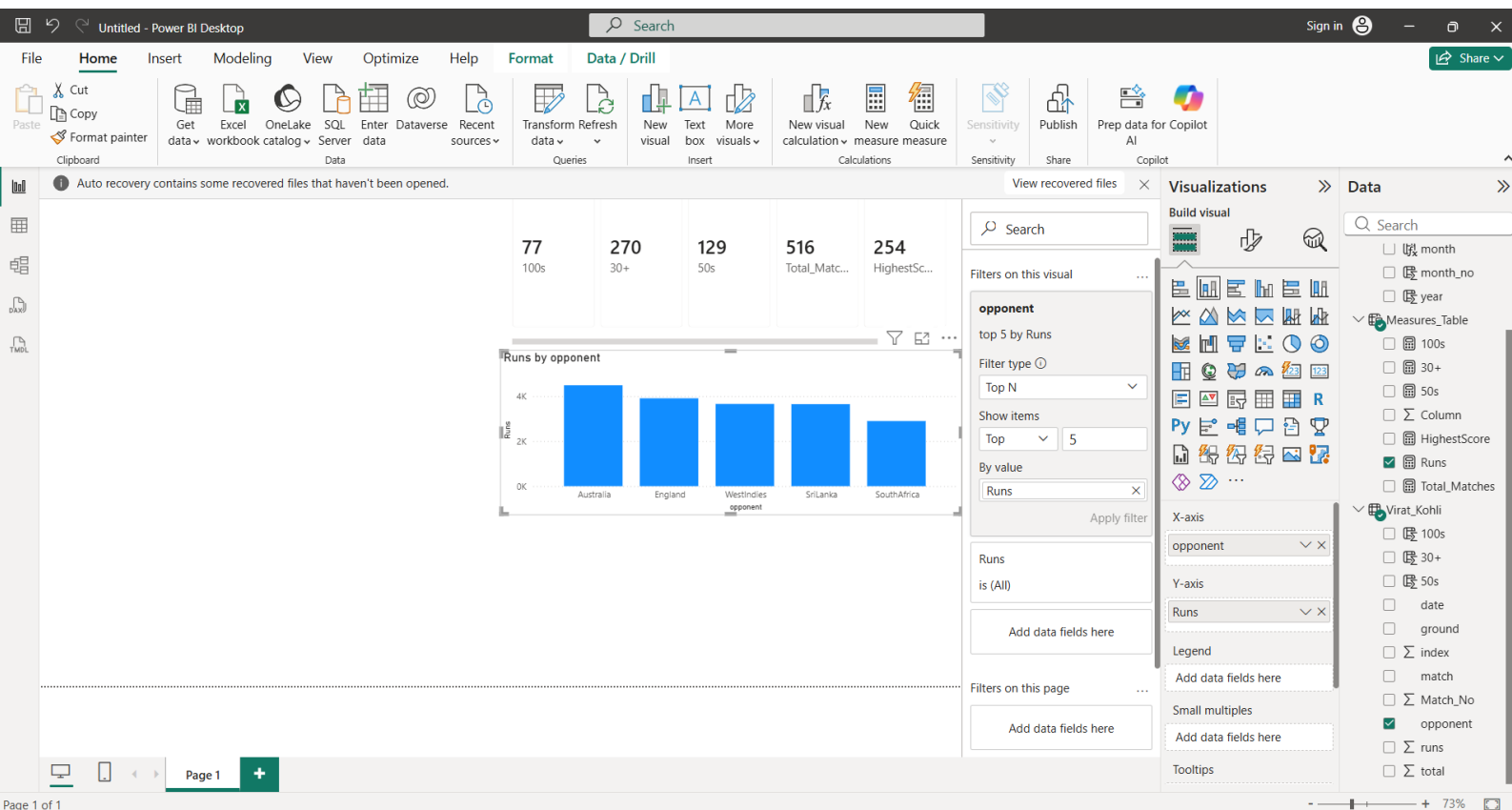
date

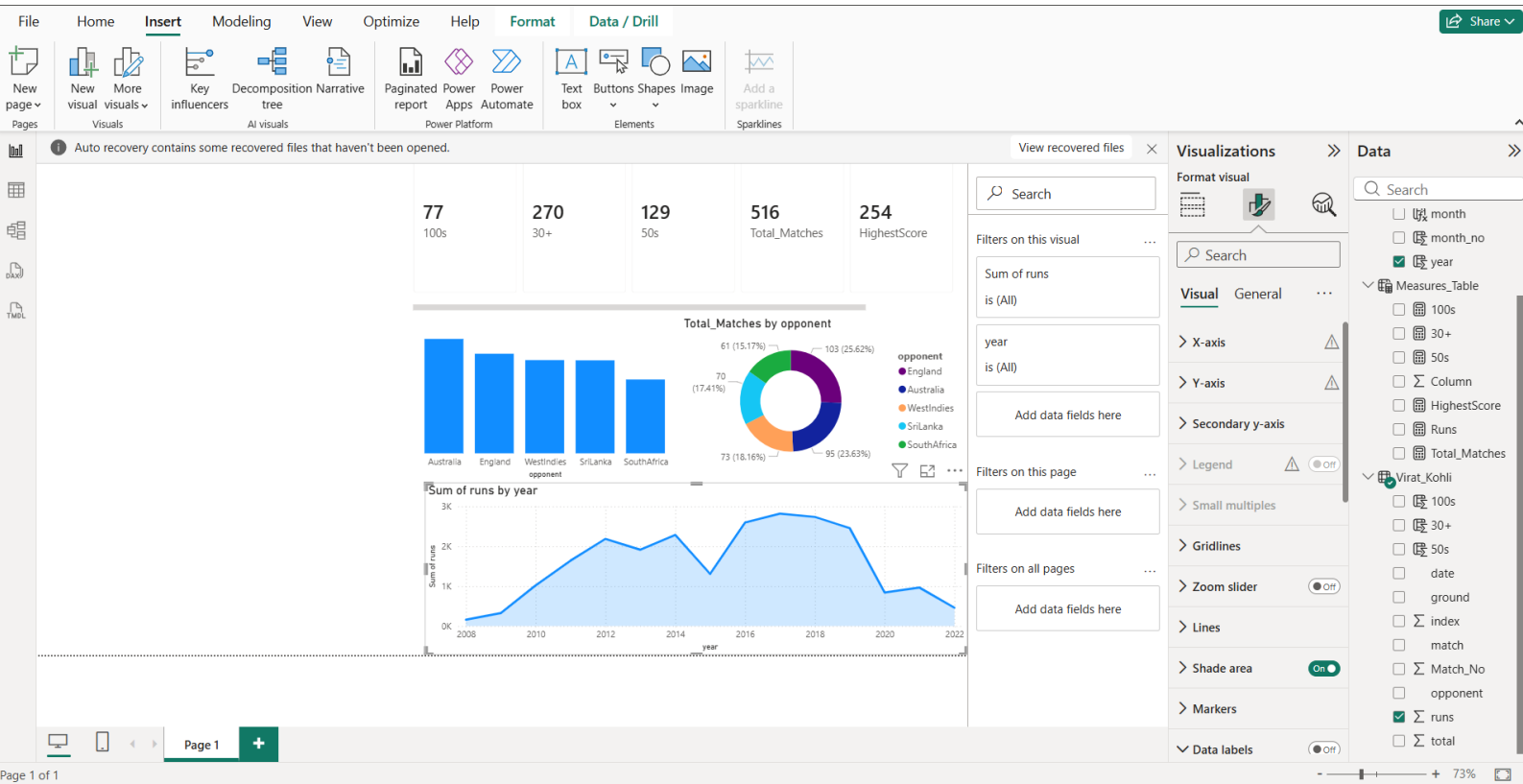
ground

index

match

Table: Measures Table (1 rows) Column: Total Matches (0 distinct values)







15

42

17

70

243

